

STOCK MARKET ANALYSIS

Historical price analysis of six Indian listed companies

Project Report

This report analyzes historical daily stock-market records supplied for Bajaj Auto, Eicher Motors, Hero MotoCorp, Infosys, TCS, and TVS Motors. It summarizes closing-price movement, data coverage, and cross-company trends using descriptive calculations. The findings describe the supplied historical period and are not investment recommendations or forecasts.

Data coverage: 01 Jan 2015 to 31 Jul 2018

Total observations: 5,334 rows across 6 company datasets.

1. Objectives

- Review daily historical closing prices for six companies.
- Calculate first-to-last closing-price change over the observed period.
- Compare the observed price ranges and visualize relative price movement.
- Prepare findings that can be reproduced in SQL and presented in a project walkthrough.

2. Methodology

Each Excel workbook was read from its company worksheet. Dates and closing prices were converted to consistent data types; rows missing either field were excluded from the calculations. Records were sorted from earliest to latest. Percentage change is calculated as $(\text{last observed close} - \text{first observed close}) \div \text{first observed close} \times 100$. This is a price-only comparison and does not include dividends, fees, taxes, or corporate-action adjustments.

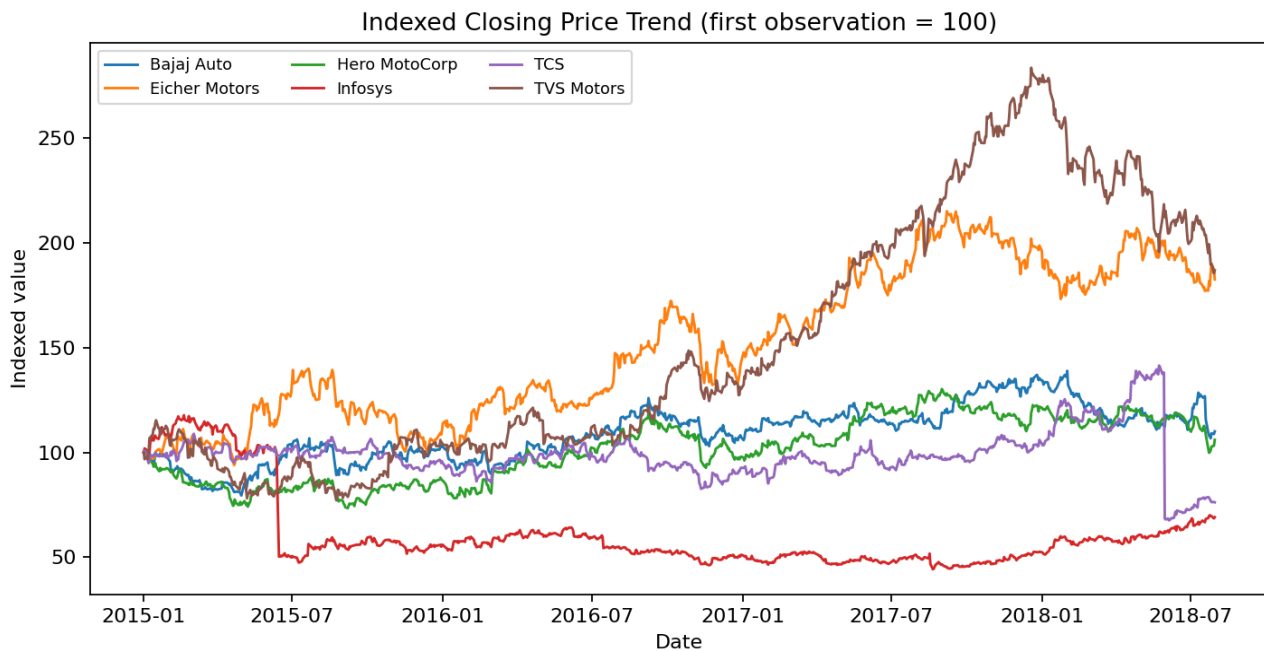
3. Company-wise summary

Company	Rows	First close (■)	Last close (■)	Change	Min close (■)	Max close (■)
TVS Motors	889	276.85	517.45	+86.91%	213.55	785.30
Eicher Motors	889	15,239.15	27,820.95	+82.56%	14,320.60	32,786.40
Bajaj Auto	889	2,454.10	2,700.70	+10.05%	1,949.40	3,409.50
Hero MotoCorp	889	3,107.30	3,293.80	+6.00%	2,279.80	4,047.30
TCS	889	2,548.20	1,941.25	-23.82%	1,721.20	3,604.80
Infosys	889	1,975.80	1,365.00	-30.91%	873.50	2,326.60

Note: figures are calculated from the uploaded workbooks as provided. The first and last observations may differ from dates mentioned in the supplied presentation, so the table uses the dates actually present in each workbook.

4. Relative closing-price trend

To compare companies with different starting share prices, each series is indexed to 100 at its first observation. Values above 100 indicate a higher closing price than the starting observation; values below 100 indicate a lower price. This indexing is descriptive and does not imply comparable risk.



5. Key observations

- The largest first-to-last percentage increase in this dataset was observed for TVS Motors (+86.91%).
- The smallest first-to-last percentage change was observed for Infosys (-30.91%).
- Closing-price ranges varied substantially across companies: for example, Eicher Motors ranged from ₹14,320.60 to ₹32,786.40 in the supplied observations.
- Closing-price movement alone is not enough to determine whether a stock should be bought or sold. A fuller assessment would consider risk, fundamentals, dividends, valuation, and the investor's time horizon.

6. SQL analysis suggestions

```
SELECT MIN(Date), MAX(Date), MIN(`Close Price`), MAX(`Close Price`) FROM company_table;  
SELECT Date, `Close Price` FROM company_table ORDER BY Date;  
SELECT AVG(`Close Price`) AS average_close FROM company_table;  
SELECT Date, `Close Price` FROM company_table ORDER BY `Close Price` DESC LIMIT 10;
```

7. Limitations

The analysis is limited to the six supplied historical datasets and their recorded fields. It does not fetch current prices or independently verify exchange data. The files appear to cover historical records ending in 2018; results should not be treated as current market information. The project should be reproduced in MySQL after importing the data and confirming the table and column names.

8. Conclusion

The project demonstrates a repeatable workflow for loading historical stock data, cleaning dates and prices, summarizing price movement, and communicating findings through tables and a chart. The same data can be imported into MySQL Workbench for SQL-based analysis and connected to the Streamlit dashboard for interactive exploration.